Contact

mercanmeh1234@gmail.com

www.linkedin.com/in/mehmet-mer (LinkedIn)

Top Skills

Internet of Things (IoT)

STM32

Presentation Skills

Languages

Turkish (Native or Bilingual)

English (Native or Bilingual)

Honors-Awards

Student Government Service Award

Berry Hill Elementary School PTA Service Award

Mehmet Mercan

CS Honors ’27 @ Purdue | ECE, Math minor | Digital Design, Embedded Software Engineering | Hackathon winner

New York City Metropolitan Area

Summary

Detail-oriented Computer Science student at Purdue University with hands-on experience in software development, automated testing, and system integration. Most interested in Digital Design and Embedded Software Engineering.

Experience

Hello World

Organizer & Board Member

January 2026 - Present (3 months)

West Lafayette, Indiana, United States

- Working towards securing $50,000+ from corporate sponsors, up from $40,000 last year.
- Inviting companies and their recruiters to interact with the competing students and share their own company's initiatives

Purdue Electric Racing

Electronics Team Memeber

September 2025 - Present (7 months)

West Lafayette, Indiana, United States

Developed embedded firmware for the team’s Formula-style electric vehicle using an STM32F4 Discovery board.

Implemented low-level control of dashboard LEDs and driver inputs using GPIO configuration and interrupt-driven routines.

Designed and tested safety-critical firmware to process pedal torque requests using ADC sampling, and signal filtering

FIRST

Alumnus, Mentor

September 2022 - Present (3 years 7 months)

Syosset, NY

FRC 9016

As a mentor:
Mentored 10 teammates in debugging, quality assurance, and system integration, contributing towards the team's continued success.

As a student:
Developed finite state machine automation in Java, leading to the #1 ranked autonomous routine in regional play.

Integrated computer vision for localization using SolvePnP, leading to the team's first Regional Championship [NYLI2, 2024].

Led team of 40+ members across 6 competitions, qualifying for the World Championships three years consecutively.

Embedded Systems @ Purdue
Embedded Software Developer | Harmonicore Project
August 2025 - December 2025 (5 months)
West Lafayette, IN

Designed and implemented FPGA-based digital signal processing (DSP) modules for real-time autotuning.

Programmed FPGA logic using SystemVerilog to realize low-latency, efficient signal transformations.

Optimized HDL modules for timing and resource efficiency, enhancing the core's responsiveness during live signal processing.

The Data Mine-Purdue University
Developer
August 2025 - December 2025 (5 months)
West Lafayette, Indiana, United States

Processed and analyzed large datasets using Python (Pandas, NumPy) and R, extracting insights to guide decisions.

Implemented high-performance computing (HPC) optimizations and containerized workflows for efficiency.

Built structured datasets via web scraping and cleaning pipelines, enabling downstream analytics.

Purdue Lunabotics
Software Developer
August 2025 - November 2025 (4 months)
West Lafayette, Indiana, United States

Implemented autonomous navigation algorithms stemming from A* to reliably traverse in simulated lunar environments.

Integrated LiDAR sensors with control systems to improve excavation accuracy and obstacle detection.

Optimized communication between software modules with ROS to ensure real-time decision making during missions.

Statice Health International
Founder, Blogging Department
January 2023 - May 2025 (2 years 5 months)
Syosset, New York, United States

Created and managed a public health blog raising awareness on conditions such as PTSD, epilepsy, and cancer.

Trained and mentored dozens of volunteer writers; edited over 20 articles for clarity, accuracy, and accessibility.

Blogs reached 150M+ total readers.

Brookhaven National Laboratory
Research Intern
July 2024 - August 2024 (2 months)
New York, United States

Designed and implemented an automated defect detection system using a camera mounted to a 3D printer platform.

Wrote Python scripts integrating OpenCV to identify silicon wafer defects critical for photolithography.

Contributed to semiconductor analysis for high-precision chip manufacturing under professional mentorship.

K12 STEM Camp
Assistant Teacher
May 2023 - August 2023 (4 months)
Melville, New York, United States

Taught elementary school (K-5) children the basics of robotics in an interactive and fun environment.

Focused on making STEM concepts—programming, engineering, and mechanics—fun and easy to understand through hands-on activities like building simple robots and programming them to complete tasks.

Education

Purdue University
Bachelor's degree, Computer Science · (December 2027)

Syosset High School